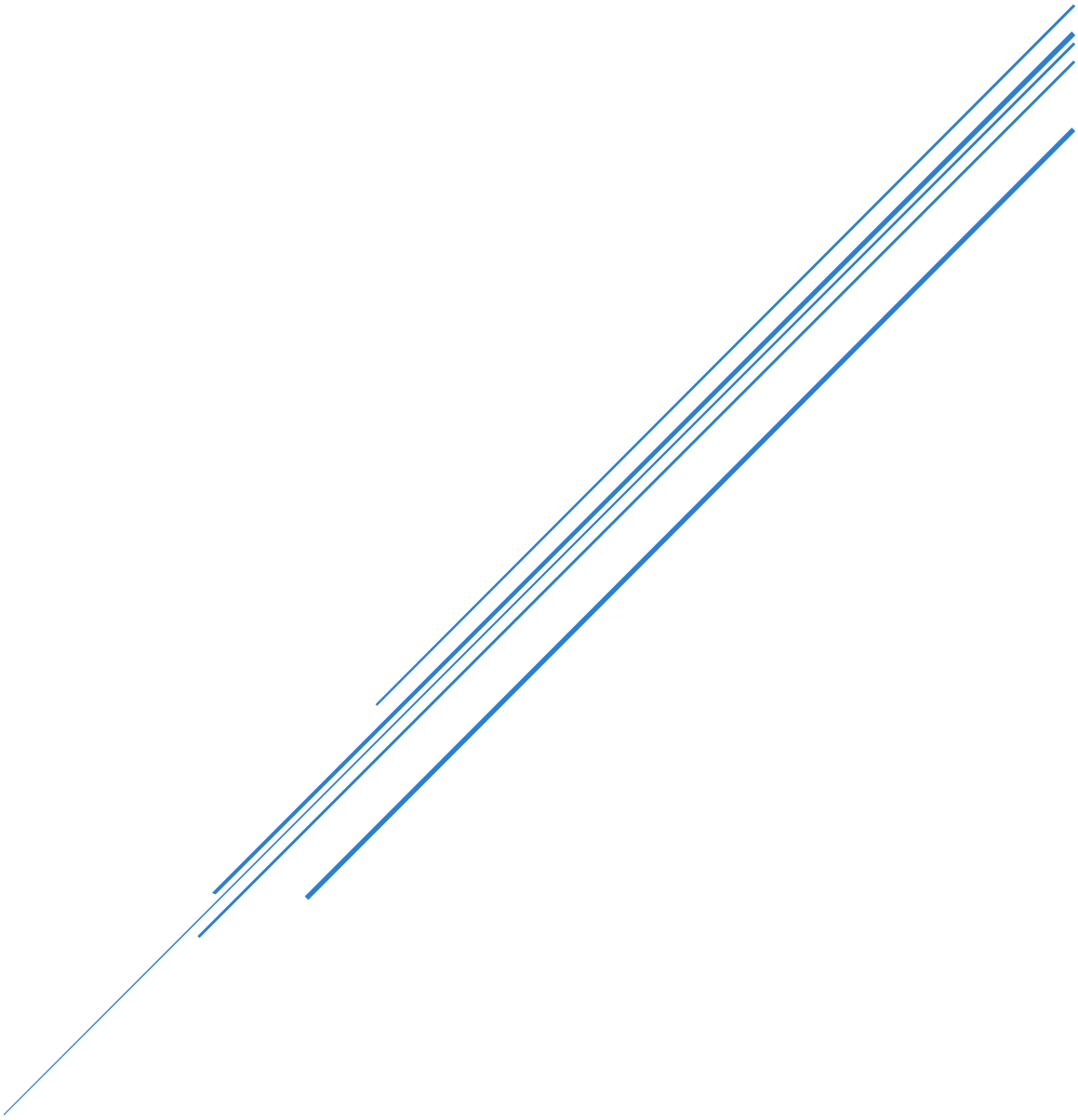
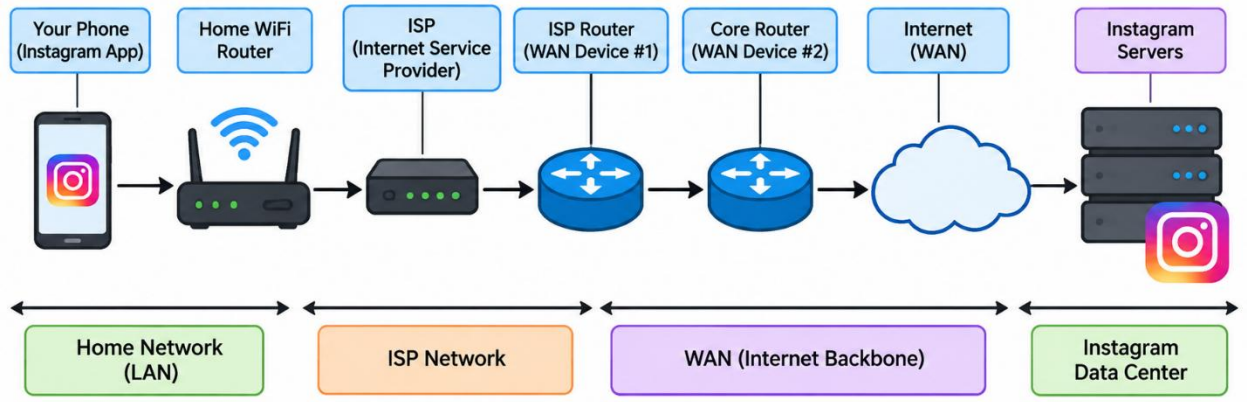
WIDE AREA NETWORKS

Assignment - 5



1. **Draw a simple diagram (on paper or using an online tool like draw.io) showing how your phone connects to Instagram's servers from your home WiFi, including the ISP and at least two Wide Area Network (WAN) devices (like routers or switches) in the path.**

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Network Diagram: Phone to Instagram Servers

The diagram shows the complete connection path from your phone to Instagram's servers through your home Wi-Fi and ISP network.

Connection sequence:

1. **Your Phone (Instagram App)**
2. **Home Wi-Fi Router**
3. **ISP (Internet Service Provider)**
4. **ISP Router (WAN Device 1)**
5. **Core Router (WAN Device 2)**
6. **Internet (WAN)**
7. **Instagram Servers**

2. **Create a table listing three types of WAN technologies (such as MPLS, Leased Line, and Broadband Internet) and compare them based on speed, cost, and where you might see them used in real life (e.g., Zomato office, IRCTC ticketing, your home).**

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WAN Technology	Speed	Cost	Real-Life Usage
MPLS	High and reliable	High	Used by large companies like Zomato to securely connect multiple office branches.
Leased Line	High and dedicated	Very High	Used by services like IRCTC for reliable connectivity between offices and data centers.
Broadband Internet	Medium to High	Low	Used in homes for browsing, online classes, YouTube, and Instagram.

3. **Research and write a short explanation of how a VPN (Virtual Private Network) uses WANs to let you access apps like Spotify or YouTube securely from a public WiFi at a coffee shop.**
   Hint: Focus on how the VPN creates a private tunnel over the public WAN.

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How VPN Uses WAN to Provide Secure Internet Access

A VPN (Virtual Private Network) creates a secure, encrypted private tunnel over a public WAN (Wide Area Network), such as the internet.

For example, when we connect our phone to public Wi-Fi at a coffee shop and open Spotify or YouTube:

1. **The phone connects to the coffee shop's public Wi-Fi.**
2. **The VPN creates an encrypted tunnel between the phone and the VPN server through the public WAN.**
3. **The data travels through this secure tunnel, making it difficult for attackers on the public Wi-Fi to read it.**
4. **The VPN server forwards the request to Spotify or YouTube.**
5. **The response returns through the VPN tunnel to the phone.**

4. **Imagine Flipkart wants to connect its warehouses in Ahmedabad, Mumbai, and Delhi. Suggest which WAN technology would be best for them and justify your answer with two reasons.**

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Best WAN Technology for Flipkart: MPLS

I would recommend MPLS (Multiprotocol Label Switching) for connecting Flipkart's warehouses in Ahmedabad, Mumbai, and Delhi.

Two Reasons:

1. High Speed and Reliability: MPLS provides fast and reliable connectivity, helping warehouses share inventory, order, and delivery information smoothly.

2. Secure Communication: MPLS supports private network connections between warehouses, helping protect important business data.